Data Exploration

Data Cleaning and Preprocessing Using Microsoft Excel

Introduction:

This project focuses on cleaning and preprocessing a product dataset using Microsoft Excel. The objective is to improve data quality by handling missing values, correcting inconsistent text formats, fixing category typos, removing duplicate records, splitting and merging columns, applying number formatting, and using conditional formatting for better visualization. The cleaned dataset ensures accuracy, consistency, and reliability for further analysis and reporting.

The dataset includes:

- Product ID
- Product Name
- Brand Name Price ($)
- Quantity
- Category

Data Analysis

1. Handling Missing Values:

● Check for missing values in the 'Price' column. How would you handle products with missing price information?

Price column :No blank values are visible Price blank formula =COUNTBLANK(D2:D35)

● If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

Category column :no blank values are visible (=COUNTBLANK(F2:F35))

Interpretation:

The dataset was checked for missing values in important columns such as Price and Category.

- Missing prices can effect calculation like average and total sales
- Missing categories can create problems during filtering and analysis. Cleaning process

• Used COUNTBLANK() to identify empty cells.

• Missing prices can be replaced with: Average price, or“0” if appropriate.

2. Correcting Inconsistent Data:

● Identify any inconsistent text formats present in the "Product Name" column.

● Identify any typos present in the "Category" column.

● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

No inconsistencies in capitalization or spelling were identified in the given records No spelling errors given data base

Interpretation:

The Product Name column was checked for inconsistent capitalization, spacing, and formatting.

Examples:

laptop

LAPTOP

Laptop

Cleaning Process

Used PROPER() to standardize text

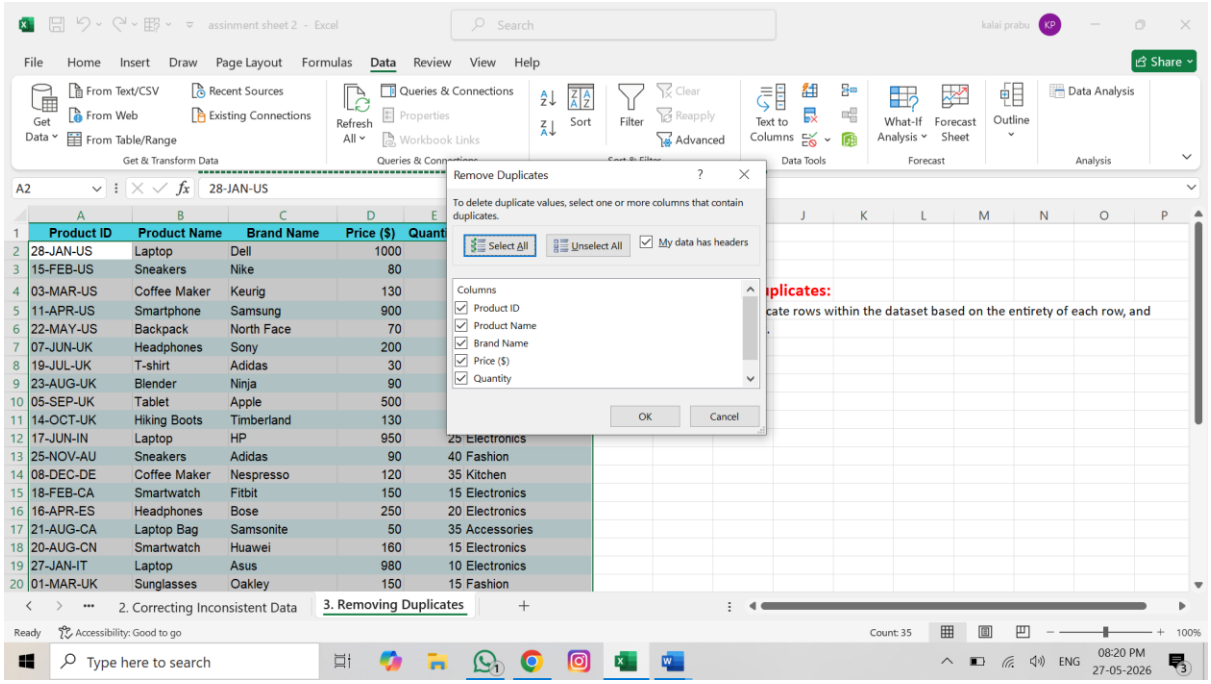
. Used TRIM() to remove extra spaces.

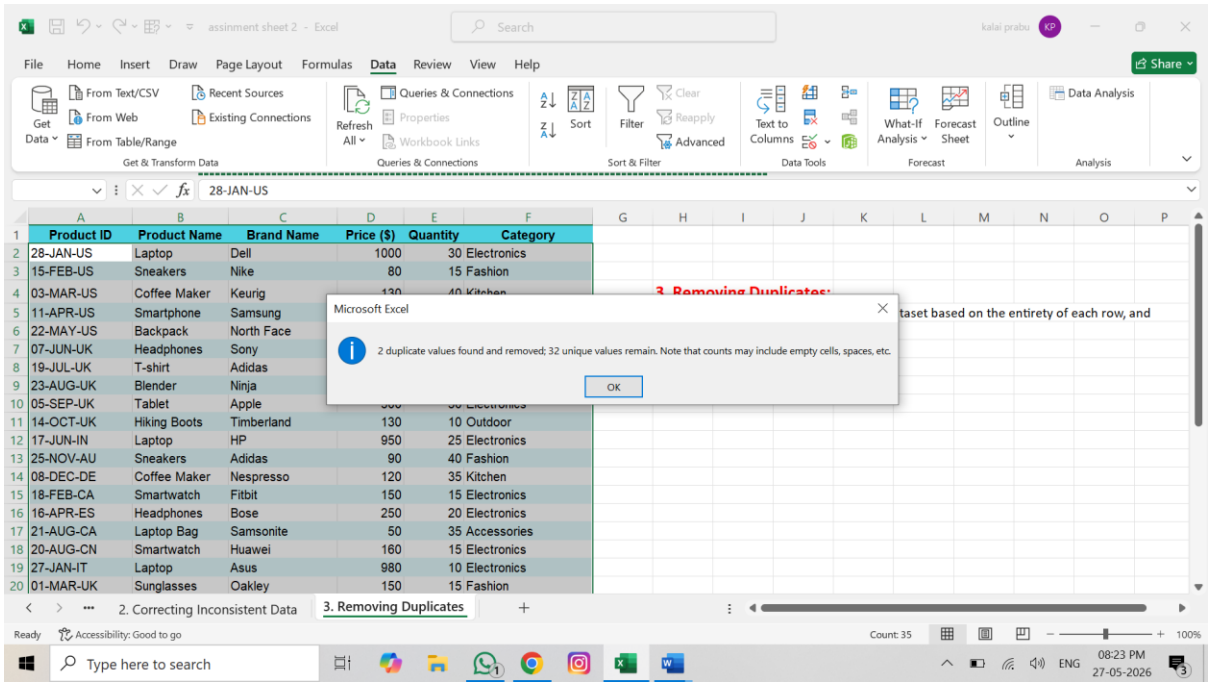
Used Find & Replace to correct spelling errors.

Formula Excel Formula =PROPER(TRIM(B2))

3. Removing Duplicates:

● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.





Interpretation:

Duplicate rows increase data redundancy and may lead to incorrect analysis results.

Cleaning Process Checked duplicate rows using all columns.

Used Excel's Remove Duplicates tool.

4. Splitting and Merging Data:

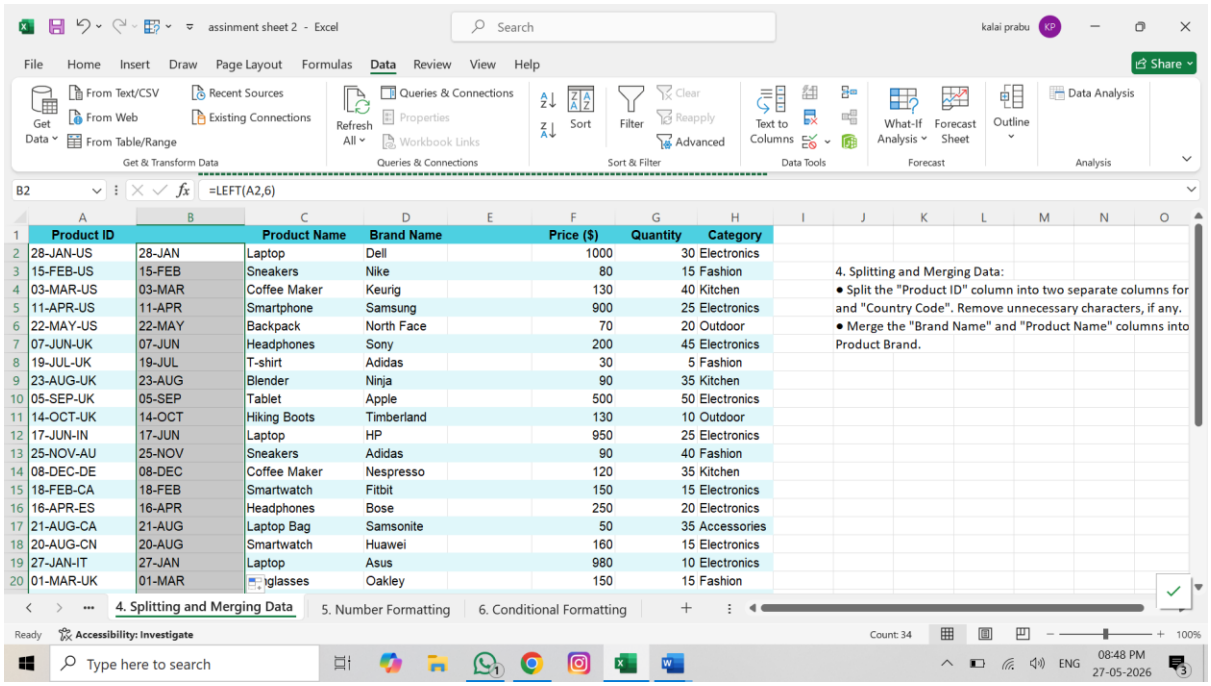
● Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code".

Remove unnecessary characters, if any.

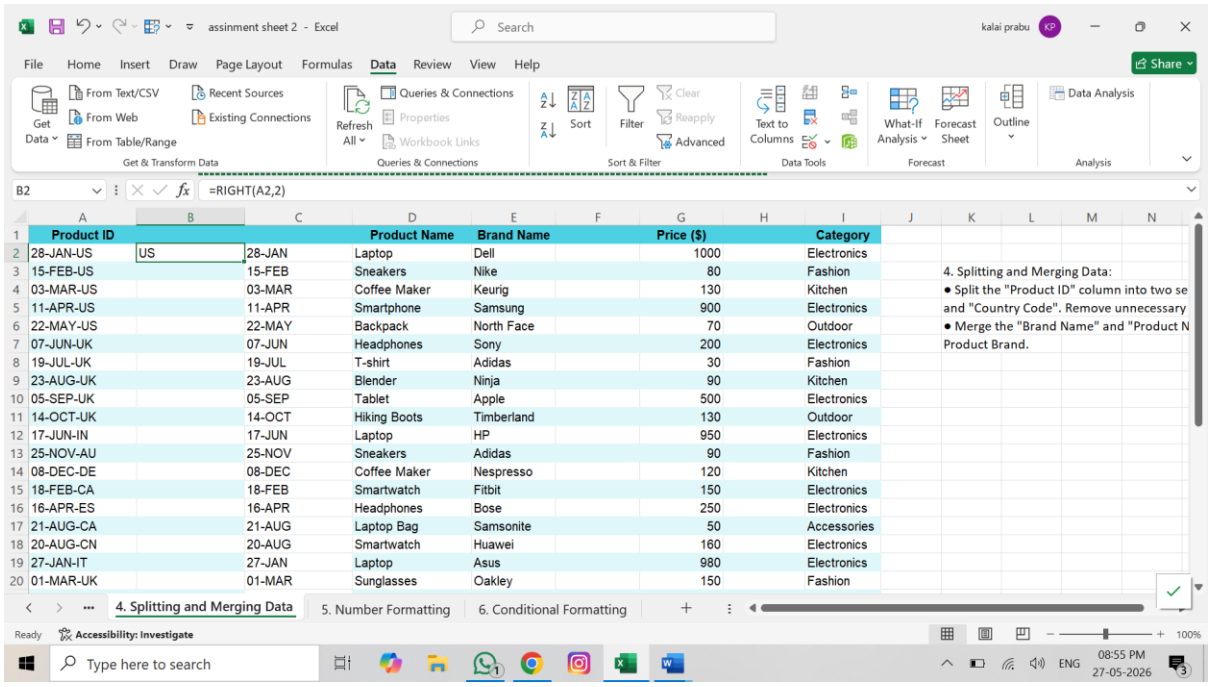
FORMULA Manufacturing date LEFT(A2,6)

Note: you double click formula not view ,I will use this format copy and paste spaical

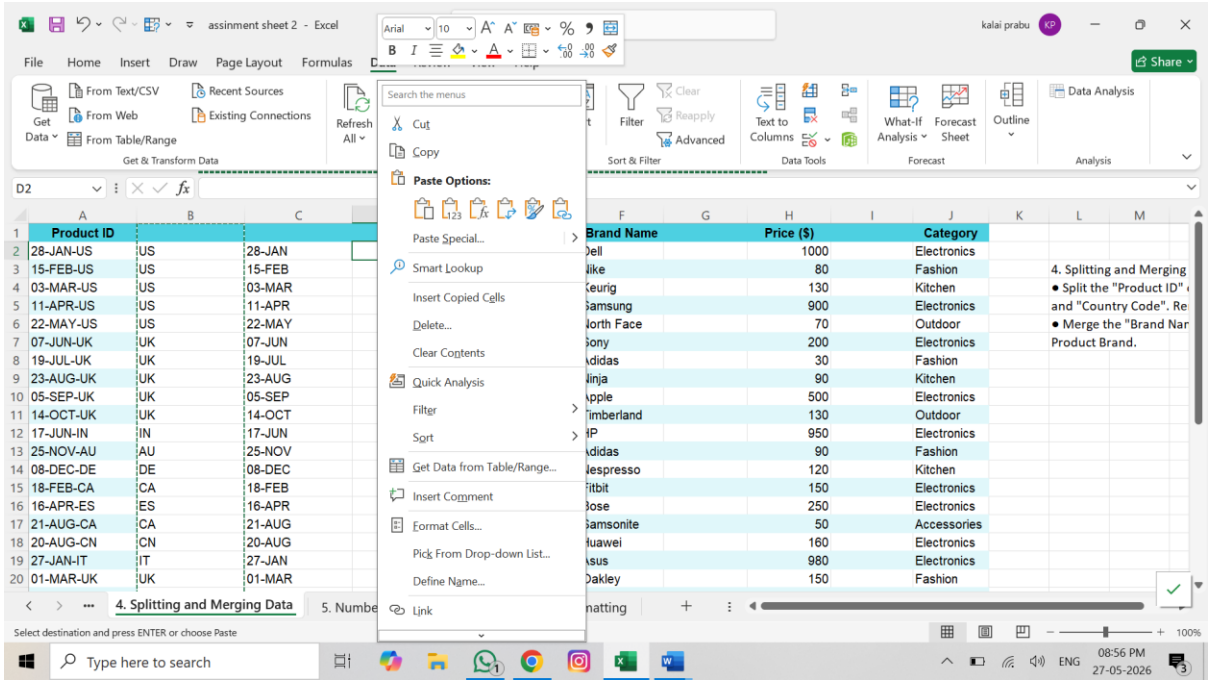
COUNTRY CODE RIGHT(A2,2)



Product ID		Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	28-JAN	Laptop	Dell	1000	30	Electronics
15-FEB-US	15-FEB	Sneakers	Nike	80	15	Fashion
03-MAR-US	03-MAR	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	11-APR	Smartphone	Samsung	900	25	Electronics
22-MAY-US	22-MAY	Backpack	North Face	70	20	Outdoor
07-JUN-UK	07-JUN	Headphones	Sony	200	45	Electronics
19-JUL-UK	19-JUL	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	23-AUG	Blender	Ninja	90	35	Kitchen
05-SEP-UK	05-SEP	Tablet	Apple	500	50	Electronics
14-OCT-UK	14-OCT	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	17-JUN	Laptop	HP	950	25	Electronics
25-NOV-AU	25-NOV	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	08-DEC	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	18-FEB	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	16-APR	Headphones	Bose	250	20	Electronics
21-AUG-CA	21-AUG	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	20-AUG	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	27-JAN	Laptop	Asus	980	10	Electronics
01-MAR-UK	01-MAR	glasses	Oakley	150	15	Fashion

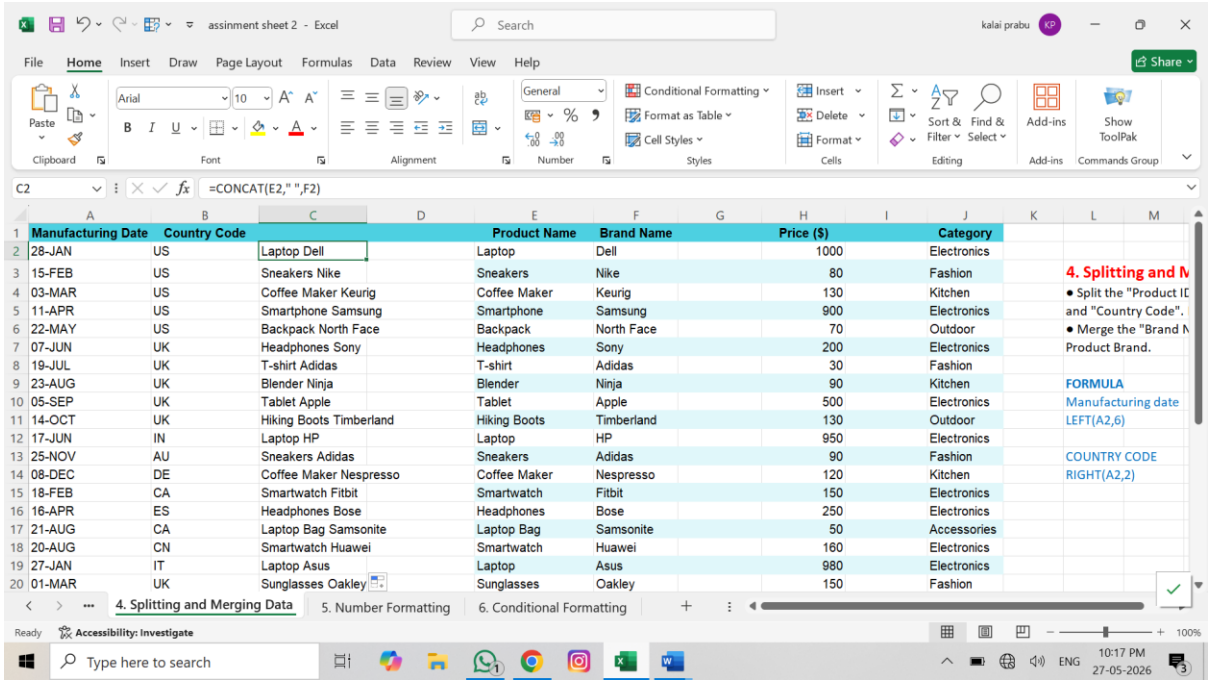


Product ID		Product Name	Brand Name	Price ($)	Category
28-JAN-US	US	28-JAN	Laptop	Dell	1000	Electronics
15-FEB-US		15-FEB	Sneakers	Nike	80	Fashion
03-MAR-US		03-MAR	Coffee Maker	Keurig	130	Kitchen
11-APR-US		11-APR	Smartphone	Samsung	900	Electronics
22-MAY-US		22-MAY	Backpack	North Face	70	Outdoor
07-JUN-UK		07-JUN	Headphones	Sony	200	Electronics
19-JUL-UK		19-JUL	T-shirt	Adidas	30	Fashion
23-AUG-UK		23-AUG	Blender	Ninja	90	Kitchen
05-SEP-UK		05-SEP	Tablet	Apple	500	Electronics
14-OCT-UK		14-OCT	Hiking Boots	Timberland	130	Outdoor
17-JUN-IN		17-JUN	Laptop	HP	950	Electronics
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08-DEC-DE		08-DEC	Coffee Maker	Nespresso	120	Kitchen
18-FEB-CA		18-FEB	Smartwatch	Fitbit	150	Electronics
16-APR-ES		16-APR	Headphones	Bose	250	Electronics
21-AUG-CA		21-AUG	Laptop Bag	Samsonite	50	Accessories
20-AUG-CN		20-AUG	Smartwatch	Huawei	160	Electronics
27-JAN-IT		27-JAN	Laptop	Asus	980	Electronics
01-MAR-UK		01-MAR	Sunglasses	Oakley	150	Fashion



● Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

Merage Brand name and Product name CONCAT(E2," ",F2)



Interpretation:

The Product ID column contained combined information such as date and country code.

Example:28-JAN-US This needed to be separated for better analysis. Cleaning Process Used LEFT() and RIGHT() functions.

Combined Brand Name and Product Name into a single column.

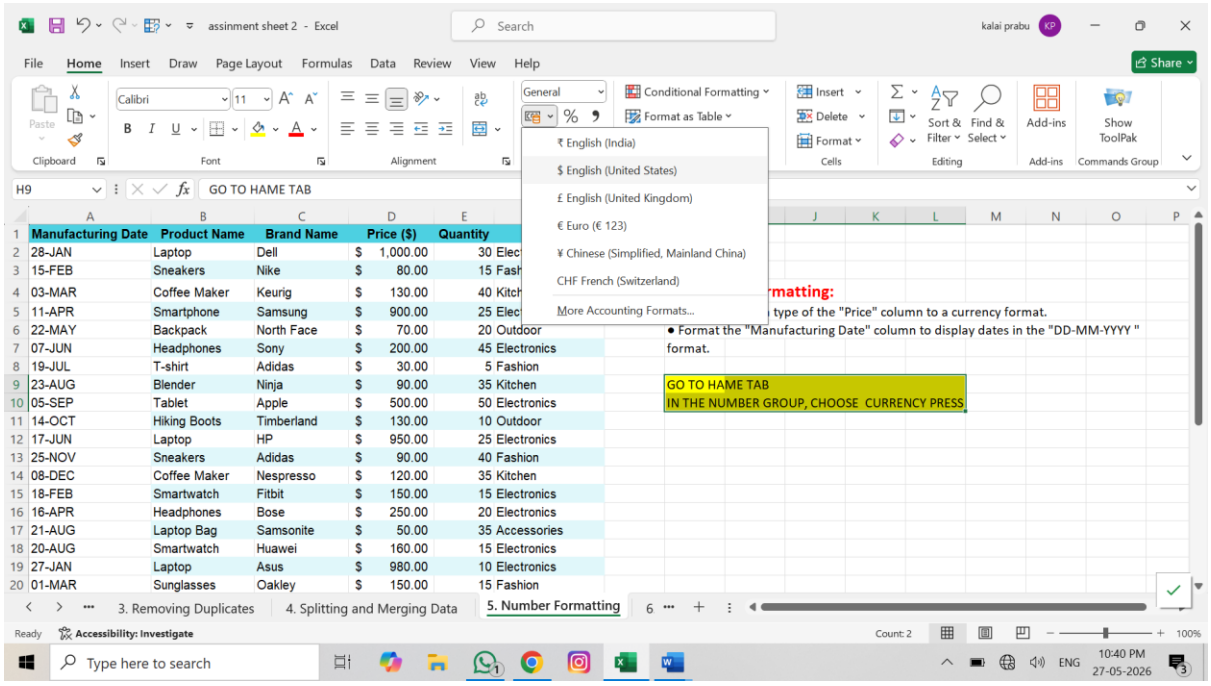
Formulas

Manufacturing Date Excel Formula =LEFT(A2,6) Country Code Excel Formula =RIGHT(A2,2)

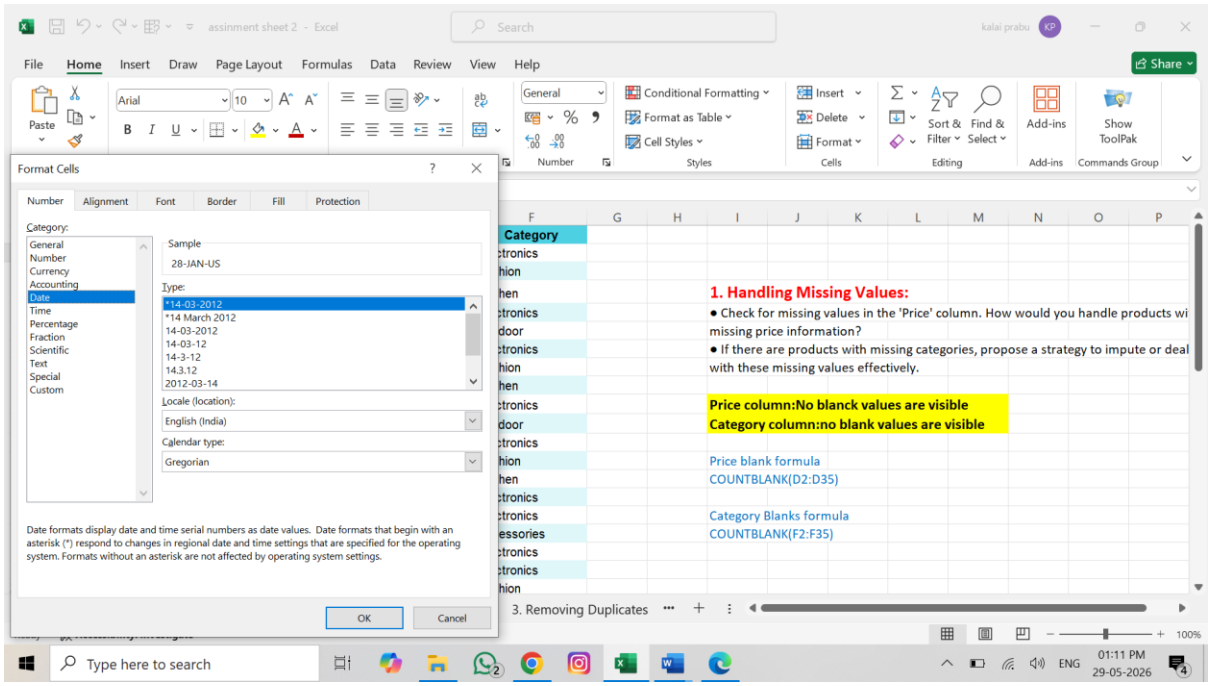
Product Brand Excel Formula:=E2&" "&F2

5. Number Formatting:

- Format the data type of the "Price" column to a currency format.



- Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.



Interpretation:

Proper formatting improves readability and makes data easier to understand. Cleaning Process

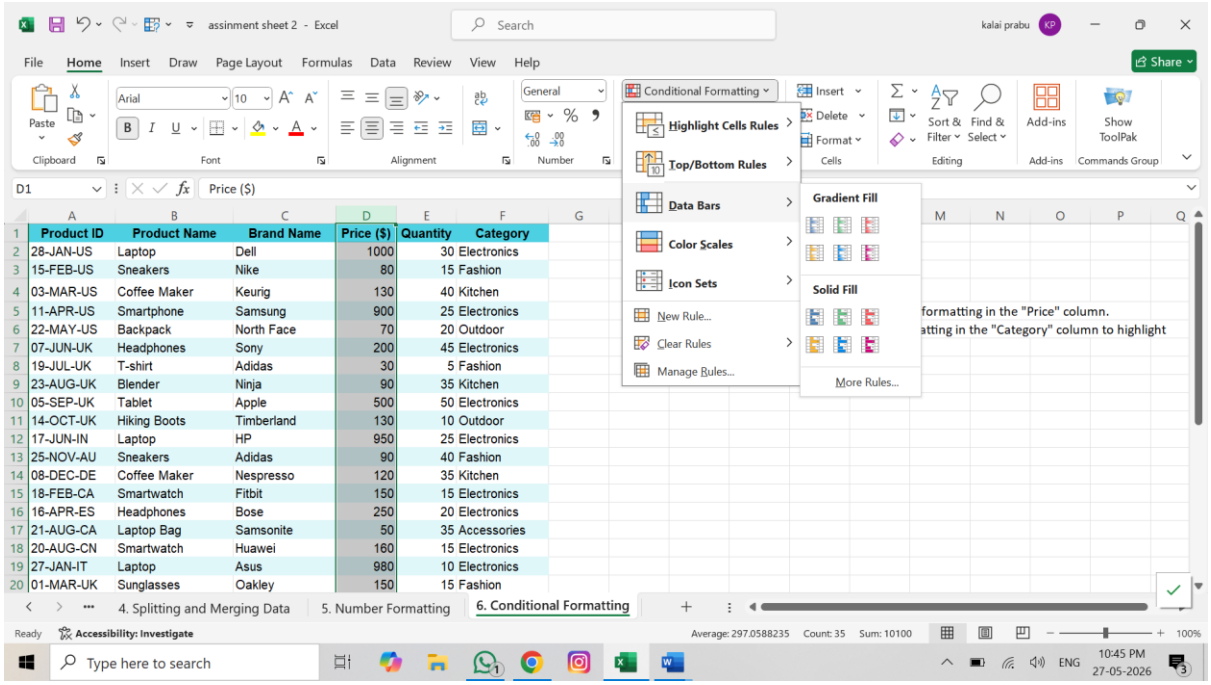
Formatted Price column as Currency.

Formatted dates into DD-MM-YYYY format.

Result Examples 1200 → $1,200.00 28-Jan-2026 → 28-01-2026.

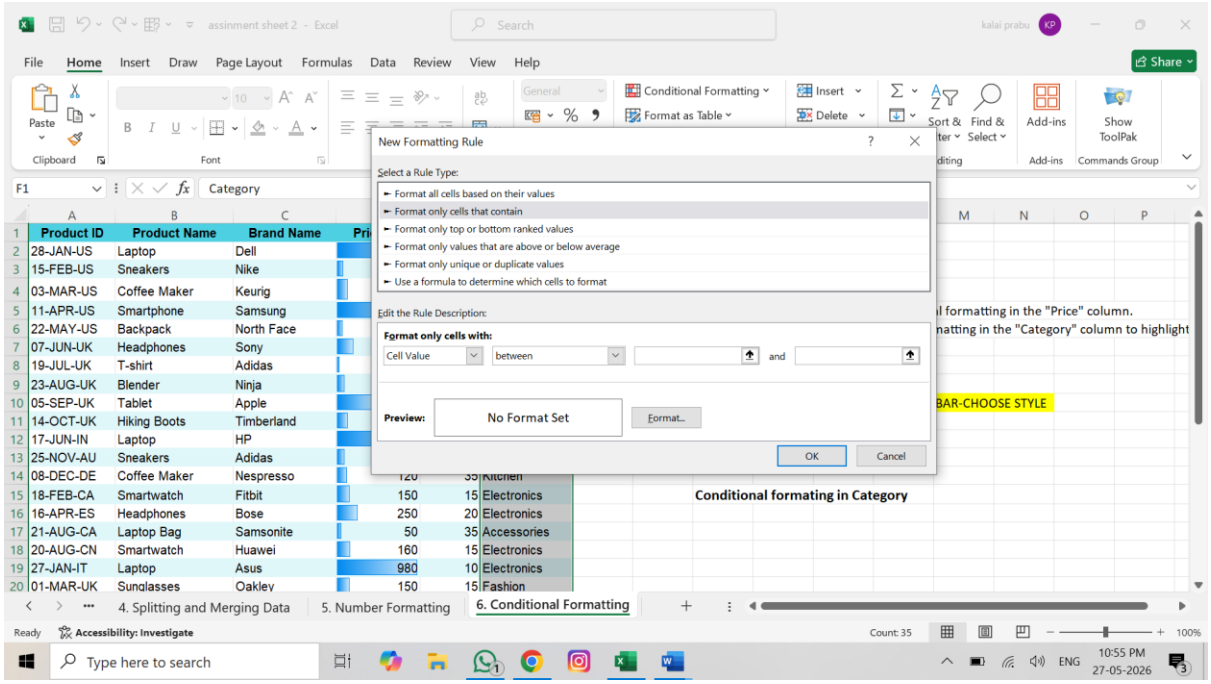
6. Conditional Formatting:

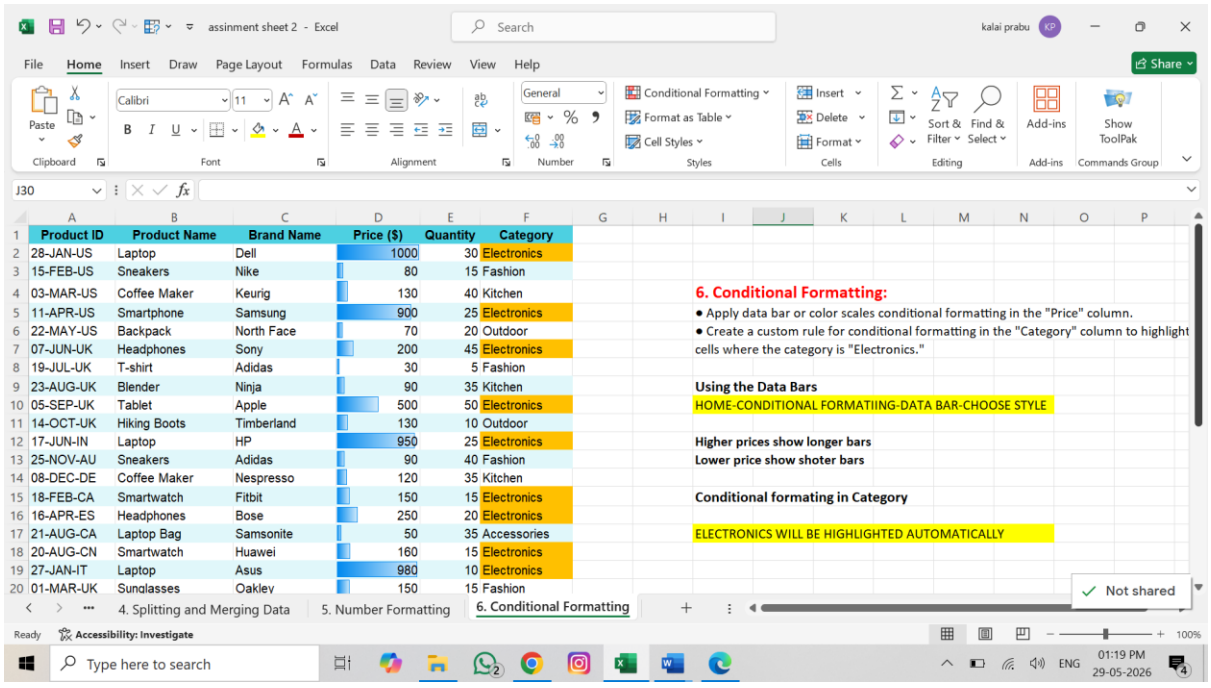
- Apply data bar or color scales conditional formatting in the "Price" column.



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27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion

- Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."





Interpretation:

Conditional formatting helps visually identify important data patterns.

Cleaning Process Applied Data Bars/Color Scales to the Price column.

Highlighted Electronics category using custom rules.

Custom Rule Plain text Cell Value = "Electronics" Result Higher prices became visually highlighted.

Electronics products were automatically colored for easy identification.

Conclusion

The dataset was cleaned through multiple processes including handling missing values, correcting inconsistent text formats, removing duplicate records, splitting and merging data, formatting numerical and date values, and applying conditional formatting. These cleaning techniques improved data accuracy, consistency, readability, and overall quality for better analysis and reporting.